MAT 504

Computer Modelling in Materials Science

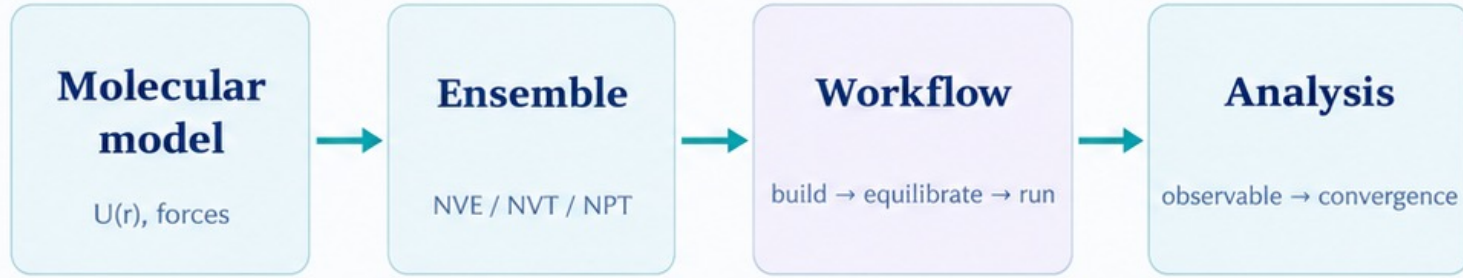
Week 4 — Molecular Dynamics in Practice

Molecular Models, Ensembles, Workflow, Analysis and Convergence

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Sabancı University

From a molecular model to a trustworthy observable



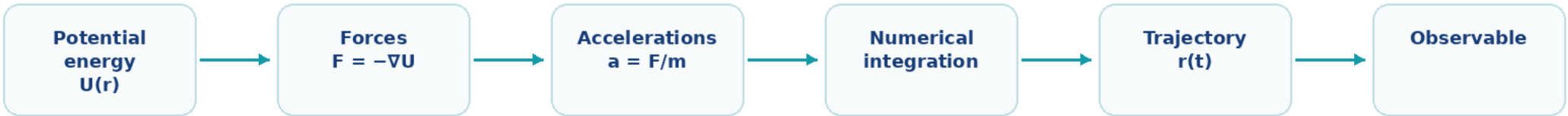
From equations of motion to reliable simulation results

WEEK 4

Where are we now?

Bridge from MD I

In Week 3, we learned how forces generate a trajectory.



Now we ask four new questions:

1. **What determines U(r) and therefore the forces?**
2. **What thermodynamic conditions should the trajectory represent?**
3. **How do we construct and run a trustworthy MD simulation?**
4. **How do we analyze the trajectory and know the result has converged?**

What does the computer need to know about the molecules?

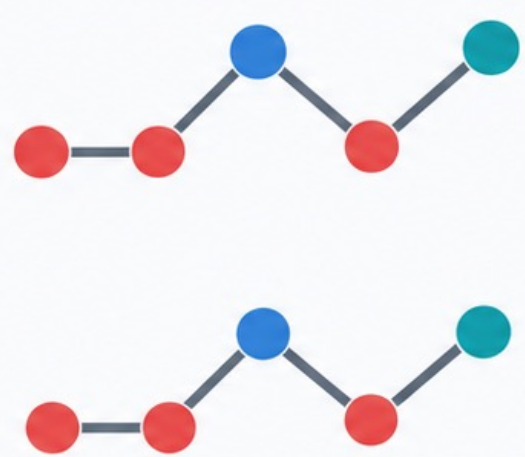
Molecular model

At every MD step, the computer must calculate the force on every atom.

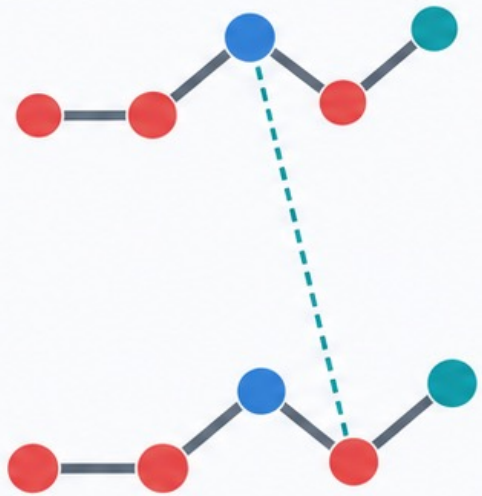
$$\mathbf{F}_i = -\nabla_i U(\mathbf{r}_1, \ldots, \mathbf{r}_N)$$

So the central modelling question is:

How do we construct U?



bonded interactions



nonbonded interactions

The force field is the physical model underlying a classical molecular-dynamics simulation.

A typical molecular-mechanics potential

Molecular model

A force field decomposes the potential energy into physically motivated contributions.

$$U = U_{\text{bond}} + U_{\text{angle}} + U_{\text{dihedral}} + U_{\text{vdW}} + U_{\text{electrostatic}}$$

Bond



bond length

Angle



bond angle

Dihedral



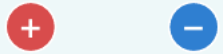
rotation

vdW



dispersion + repulsion

Electrostatic



charge–charge

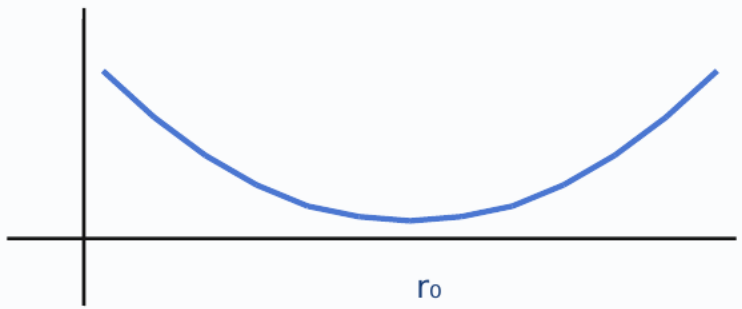
The functional form tells us how energy depends on geometry; parameters determine the strength and preferred geometry of each interaction.

Bonded interactions

Molecular model

Bond stretching

$$U_{\text{bond}} = \frac{1}{2} k_b (r - r_0)^2$$



Angle bending

$$U_{\text{angle}} = \frac{1}{2} k_\theta (\theta - \theta_0)^2$$



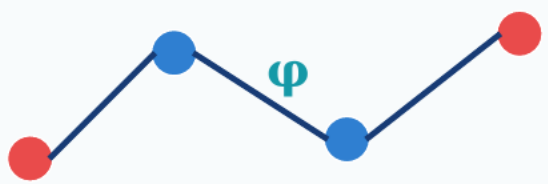
These terms keep local molecular geometry near preferred values.

Torsional interactions

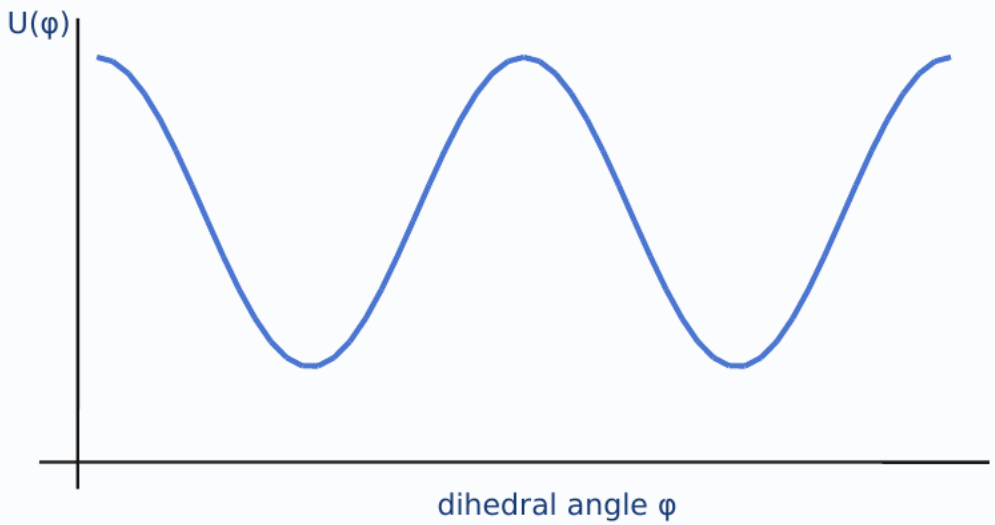
Molecular model

Rotation about a bond changes molecular conformation and therefore potential energy.

$$U_{\text{dihedral}} = k_\phi[1 + \cos(n\phi - \delta)]$$



Different φ → different conformations



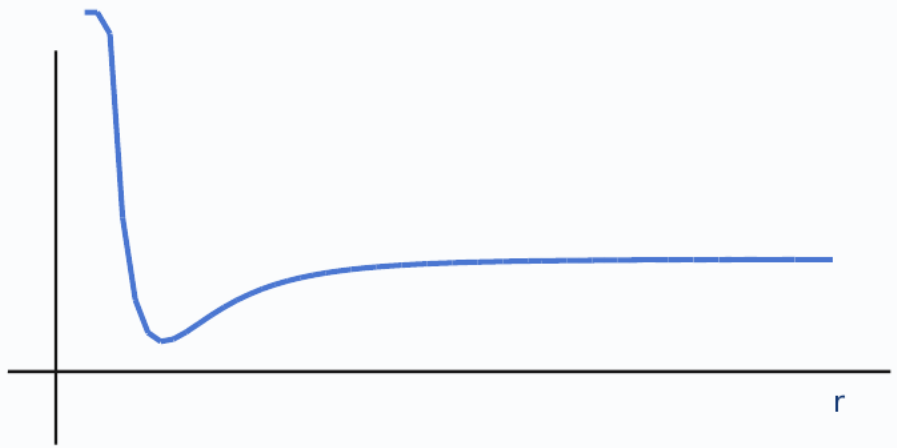
Torsional terms encode which conformations are energetically preferred.

Nonbonded interactions

Molecular model

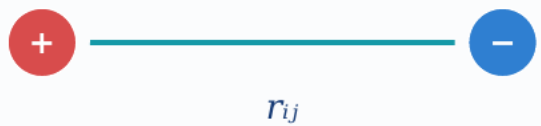
Lennard–Jones

$$U_{\mathrm{LJ}}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$



Electrostatics

$$U_{\mathrm{Coulomb}} = \frac{1}{4\pi\epsilon_0}\frac{q_i q_j}{r_{ij}}$$



r_{ij}

In condensed-phase simulations, evaluating nonbonded interactions often dominates computational cost.

What is a force field?

Molecular model

A force field is more than a list of equations.

Functional form

How energy depends on geometry and interactions

+

Parameters

Equilibrium values, force constants,
charges, ε, σ, …

=

Force field

The physical model used
by classical MD

Examples: AMBER CHARMM OPLS GROMOS

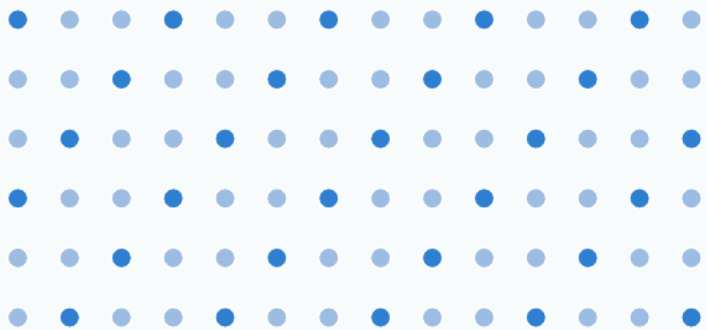
Different force fields are different approximations; model choice can change the predicted behavior.

The finite-system problem

Bulk matter

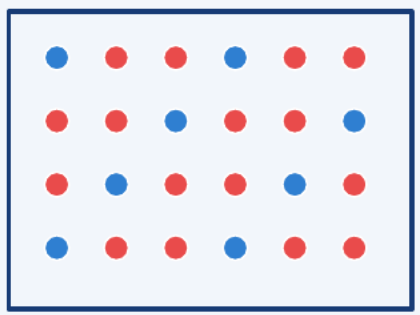
Real materials are enormous. Our MD simulation contains only a finite number of atoms.

Real condensed matter



~10^{23} particles

Simulation cell



10^{3}–10^{6} particles

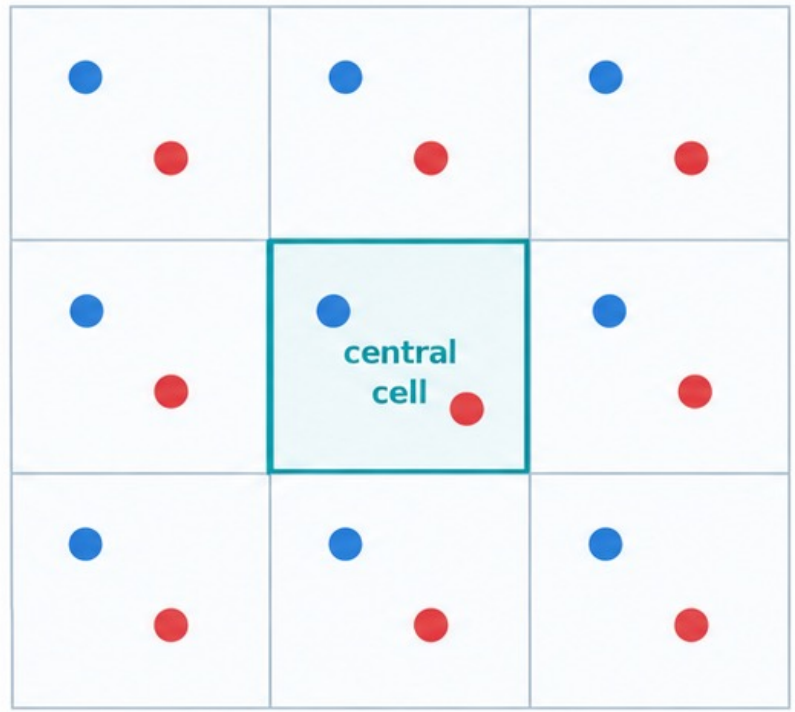
We need boundary conditions and approximations that allow a small box to represent bulk behavior.

Periodic boundary conditions

Bulk matter

A finite simulation cell is surrounded by periodic copies of itself.

- No physical walls
- Particle leaving one side re-enters through the opposite side
- Bulk-like environment



Important

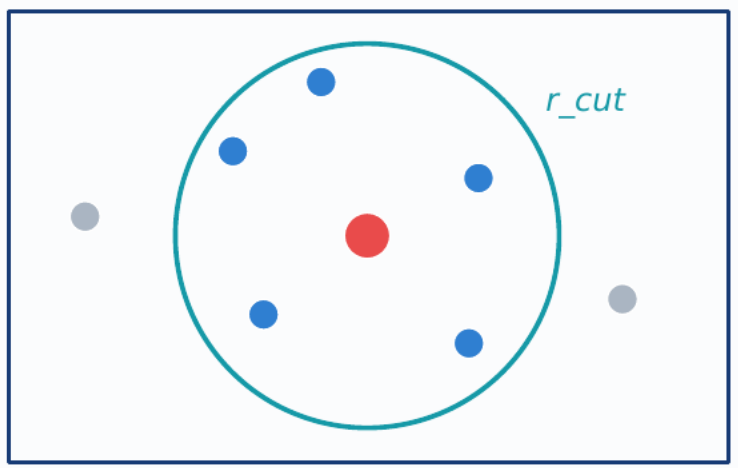
PBC remove surfaces, but they do not eliminate finite-size effects.

Periodic boundaries make the local environment repeat; they do not make the simulated system truly infinite.

Minimum image and interaction cutoffs

Computational approximations

Naïvely evaluating every pair interaction scales approximately as $O(N^2)$.



Only nearby pairs are evaluated explicitly

Minimum-image convention

For periodic systems, use the nearest periodic image of each interaction partner.



shortest periodic separation

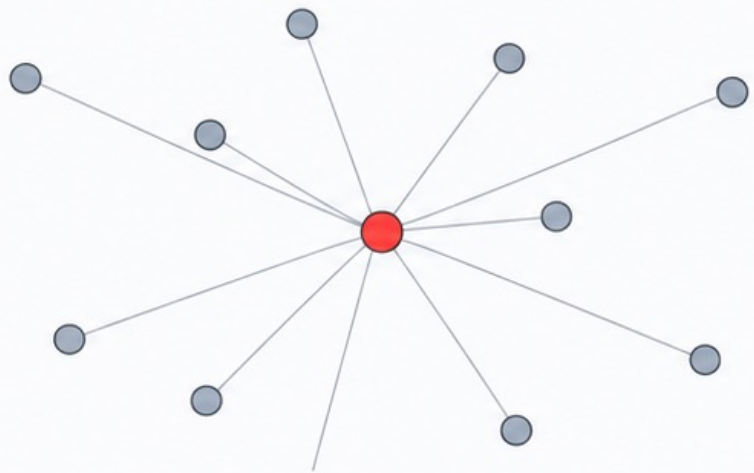
Cutoffs reduce cost, but the approximation must respect the physics of the interaction.

Neighbor lists: avoid searching the whole system every step

Computational approximations

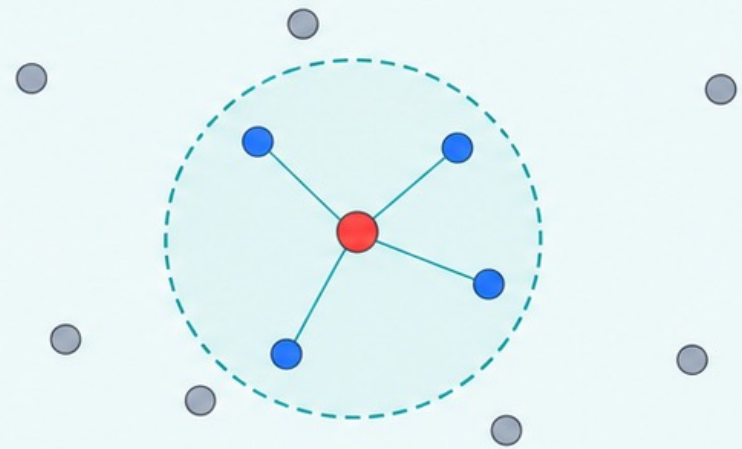
A simple idea can save a large amount of computation.

Without a neighbor list



check many pairs

With a neighbor list



store nearby partners and update periodically

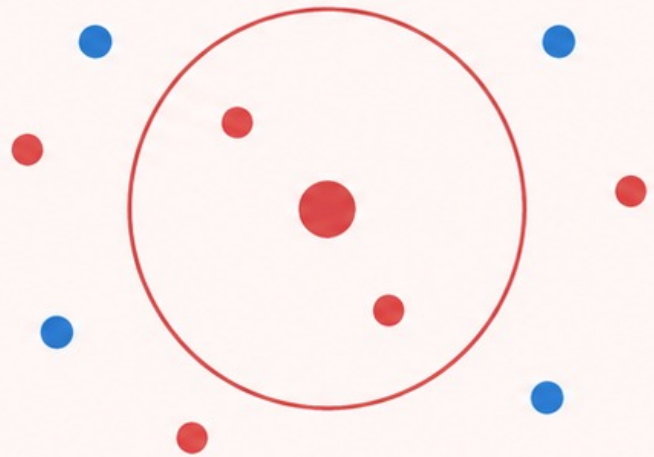
Algorithm design is part of molecular simulation: efficiency determines which systems and timescales are accessible.

Long-range electrostatics are different

Computational approximations

Coulomb interactions decay slowly with distance, so simply truncating them can be problematic.

Naive cutoff



Ewald / Particle-Mesh Ewald (PME)

Conceptually:

short range in real space + **long range handled efficiently**

No derivation needed here — the important point is that long-range interactions require special treatment.

The numerical treatment must match the range and character of the physical interaction.

From statistical mechanics to molecular dynamics

Ensembles

Week 2: equilibrium probability distribution

$$P(x) \propto e^{-\beta U(x)}$$

Monte Carlo

Generate configurations with the correct equilibrium weights

→

Molecular dynamics

Generate a trajectory by integrating equations of motion

New question: what thermodynamic ensemble does that trajectory represent?

NVE: the natural ensemble of conservative MD

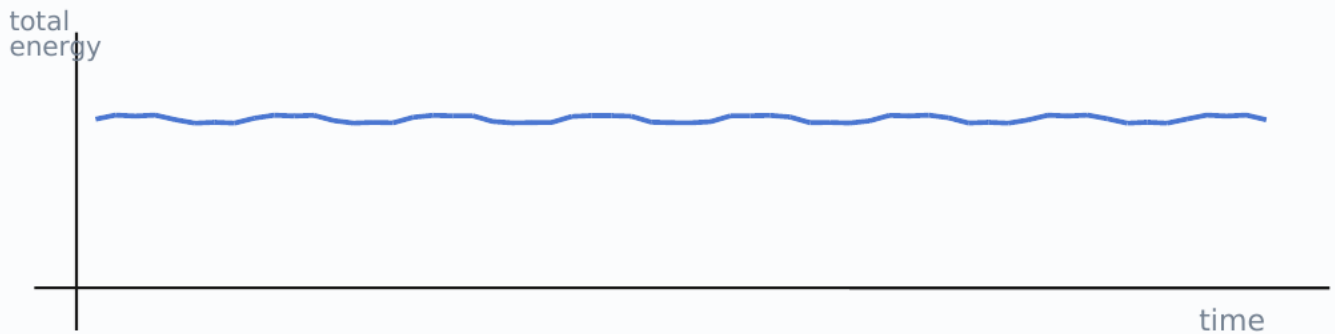
Ensembles

NVE

N, V, E constant

For an isolated classical system with a conservative potential:

$$E = K + U$$ should remain approximately constant



Energy conservation is both a physical property of NVE dynamics and a useful numerical diagnostic.

NVT: sampling at a chosen temperature

Ensembles

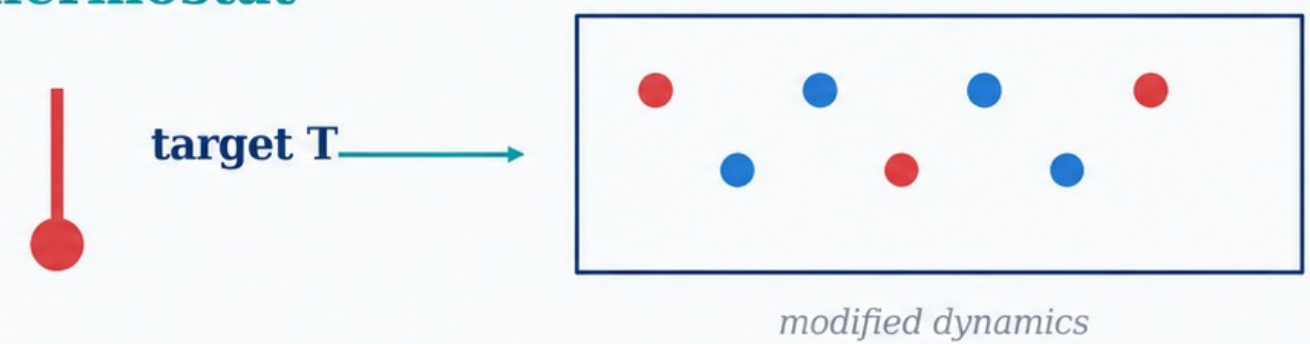
NVT

N, V, T controlled

Temperature is connected to kinetic energy.

$$K = \sum_i \frac{1}{2} m_i v_i^2$$

Thermostat



modified dynamics

A thermostat does not merely "set T = 300 K"; it modifies the dynamics to sample a chosen thermal environment.

Thermostats: different ways to control temperature

Ensembles

Different thermostat algorithms couple the system to a thermal bath in different ways.

Velocity rescaling

adjust kinetic energy directly

Andersen

random collisions with a heat bath

Langevin

friction + random force

Nosé–Hoover

extended deterministic dynamics

For now, remember the purpose — not the detailed algorithms.

NPT: allowing the volume to respond to pressure

Ensembles

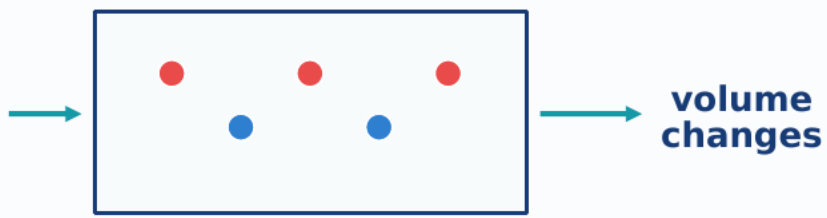
NPT

N, P, T controlled

Why use NPT?

- Many experiments are performed at specified temperature and pressure.
- Density and volume should be allowed to adjust.

Barostat



A barostat changes the simulation volume to generate the desired pressure distribution.

Which ensemble should I use?

Ensembles

There is no universally “best” ensemble — the choice follows from the physical question.

NVE

isolated system

energy-conserving dynamics and numerical diagnostics

NVT

fixed volume + temperature

equilibrium at chosen T when V is prescribed

NPT

fixed pressure + temperature

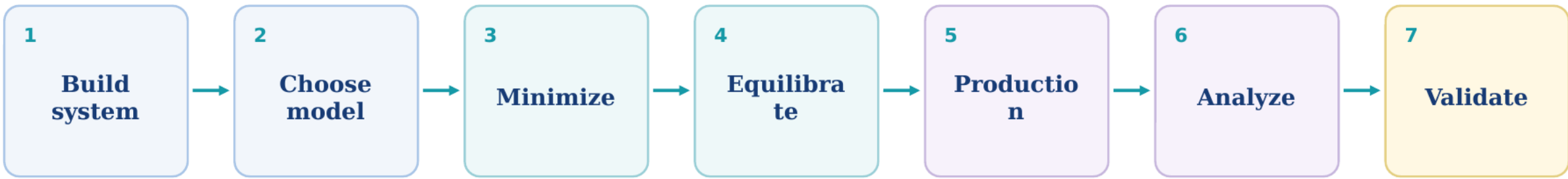
equilibrium when density / volume should adjust

Choose the ensemble to represent the thermodynamic constraints relevant to the system and observable.

An MD simulation is more than pressing “Run”

Practical workflow

A reliable simulation is a sequence of modelling decisions and preparation steps.



Every stage can influence the final observable.

A trajectory is not automatically a scientific result; the protocol determines whether it is meaningful.

Step 1 — Build the system

Practical workflow

Define the physical system you actually intend to simulate.

Coordinates

initial geometry

Composition

molecules, ions, solvent

Simulation box

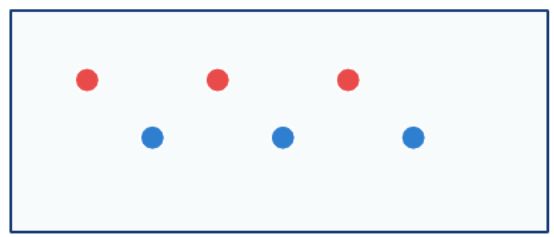
size and shape

Force-field parameters

interaction model

Boundary conditions

usually periodic for bulk



The initial coordinates are only a starting guess — they are not yet an equilibrated physical state.

Step 2 — Energy minimization

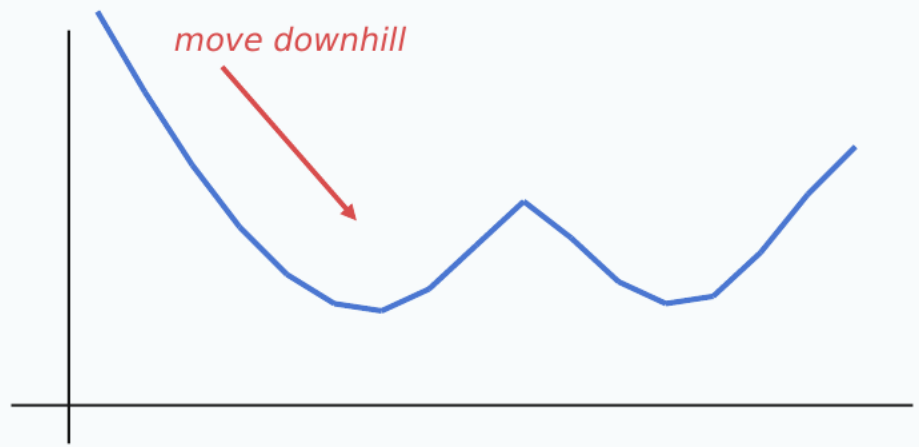
Practical workflow

Purpose: remove severe overlaps and highly unfavorable contacts before dynamics.

Before minimization



Energy landscape picture



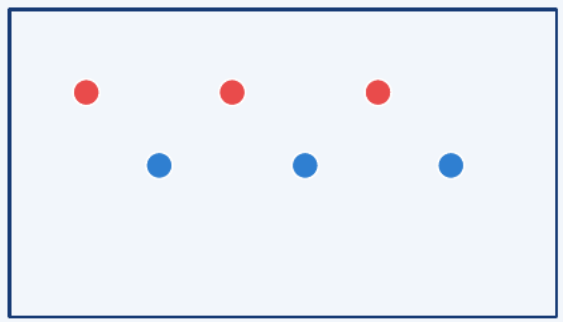
Minimization is not molecular dynamics.

Step 3 — Assign initial velocities

Practical workflow

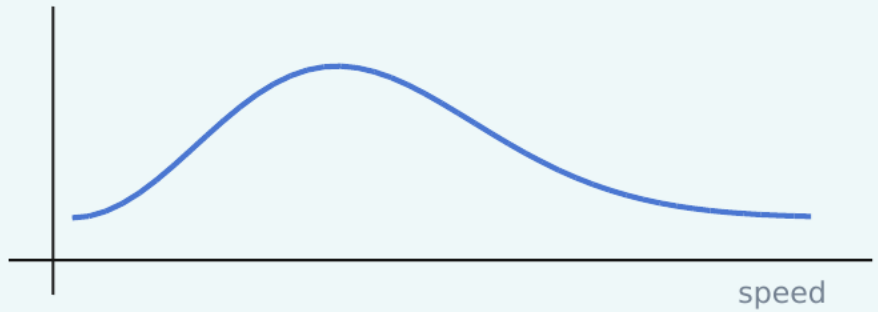
A trajectory needs both positions and velocities.

Initial positions



Initial velocities

commonly drawn from a Maxwell–Boltzmann distribution



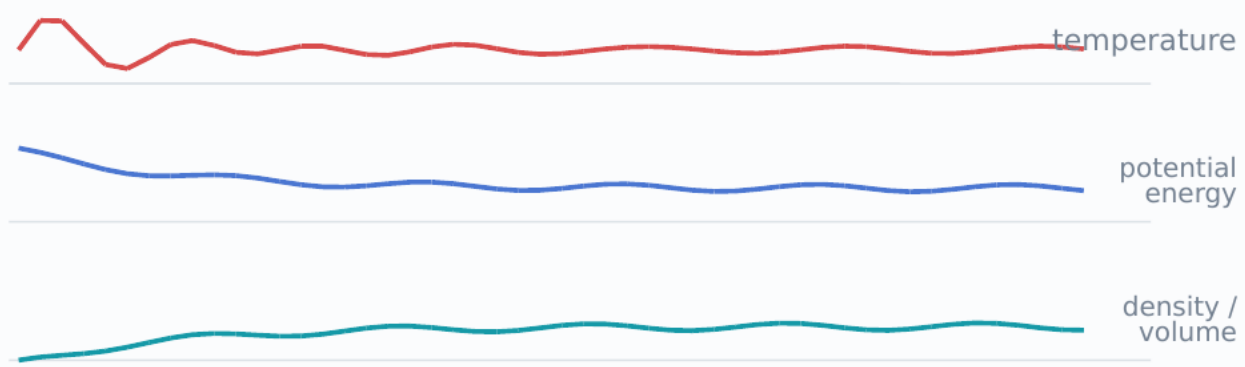
Initial velocities determine the initial kinetic energy and are chosen to be consistent with the target temperature.

Step 4 — Equilibration

Practical workflow

Has the system forgotten its arbitrary starting configuration?

Monitor key observables



Do not rely on only one curve.

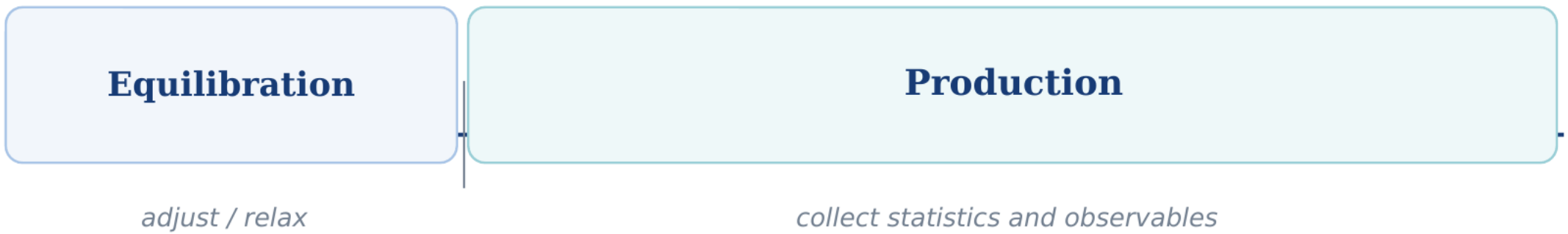
Equilibration should be judged using observables relevant to the system and question.

Equilibration is about loss of memory of the starting state, not simply about a flat-looking trace.

Step 5 — Production simulation

Practical workflow

Only after equilibration do we collect the trajectory intended for scientific analysis.



Equilibration trajectory ≠ production trajectory

How long should a simulation be?

Practical workflow

There is no universal answer such as “100 ns”.

Fast local motion

bond vibrations
local rearrangements

shorter timescales

Diffusion / relaxation

transport
collective motion

intermediate timescales

Rare transitions

folding
nucleation / barrier crossing

can require very long times

Run long enough to sample the processes that control the observable — and check convergence.

Choosing the time step

Practical workflow

The time step balances numerical accuracy against computational cost.

Too small

accurate
but expensive

many integration steps

Balanced

resolves fastest
relevant motion

stable + efficient

Too large

integration error
energy drift /
instability

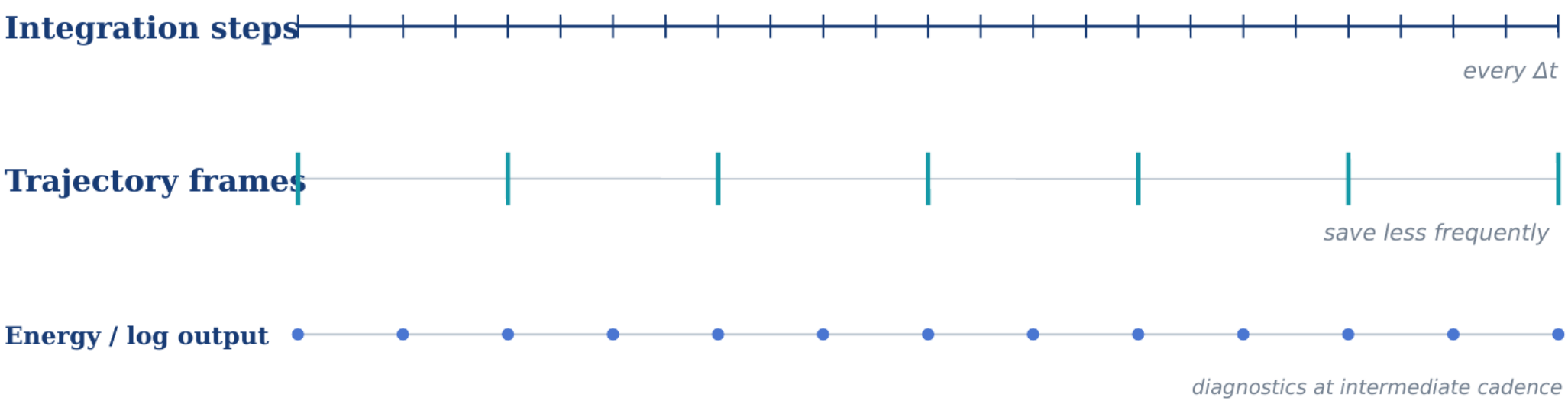
unreliable trajectory

Constraints on very fast bond vibrations can permit a larger Δt.

What should I save?

Practical workflow

Integration and trajectory output are not the same thing.



Saving every integration step is usually unnecessary and can create enormous trajectory files.

Before trusting an MD simulation

Reliability

A technically completed simulation is not automatically a trustworthy simulation.

- ✓ **Energy behavior** — appropriate for the chosen ensemble
- ✓ **Temperature / pressure / density** — consistent with the intended state
- ✓ **Equilibration** — initial-condition memory removed
- ✓ **Structural sanity** — no obvious artifacts
- ✓ **Sampling / convergence** — observable is stable with more data
- ✓ **Sensitivity / reproducibility** — result is not a fragile numerical accident

Trust comes from diagnostics, convergence, and physical consistency — not from the fact that the code ran without error.

Molecular Dynamics IV: Analysis and Convergence

Analysis & convergence

A trajectory is data. The scientific question is what we can infer from it — and how confidently.



Key idea

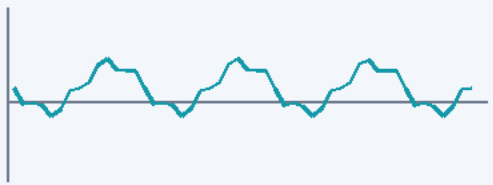
A simulation is not converged merely because it ran without crashing or because one curve looks flat.

What can we measure from an MD trajectory?

Analysis & convergence

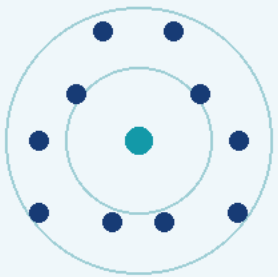
Different observables ask different questions of the same trajectory.

Thermodynamic / scalar



- energy
- temperature
- pressure
- density

Structural



- radial distribution function
- coordination / contacts
- distances, angles, RMSD

Dynamical



- mean-squared displacement
- diffusion coefficient
- time-correlation functions

Convergence must be judged for the observable you actually care about.

From ensemble averages to time averages

Analysis & convergence

In equilibrium MD, we often estimate an ensemble average from a sufficiently sampled trajectory.

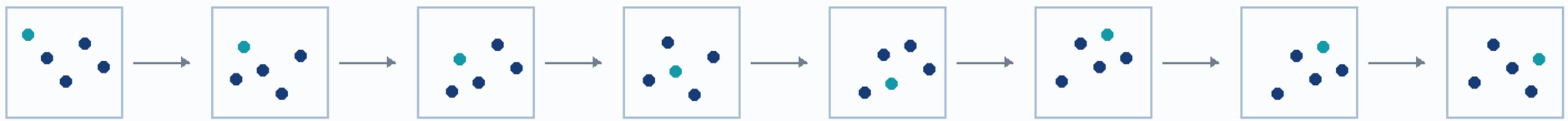
Ensemble average

$$\langle A \rangle = \int A(x)\, P(x)\, dx$$

ergodic sampling →

Trajectory estimate

$$\bar{A} \approx (1/M) \sum_k A(t_k)$$



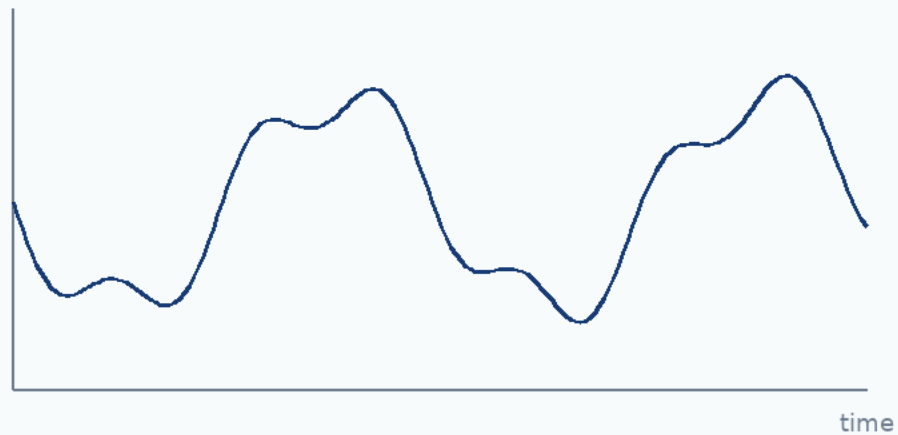
The equality is useful only if the trajectory explores the relevant equilibrium states.

Trajectory frames are correlated

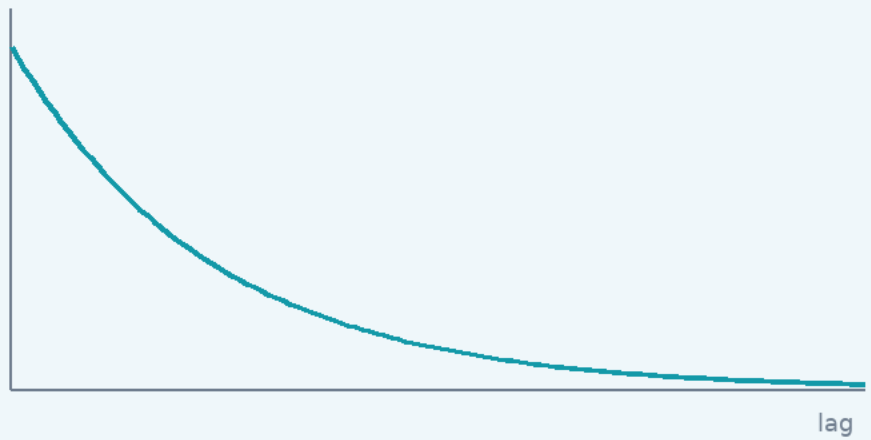
Analysis & convergence

Saving more frames does not automatically create more independent information.

Observable A(t)



Autocorrelation C(k)



100,000 saved frames ≠ 100,000 independent samples

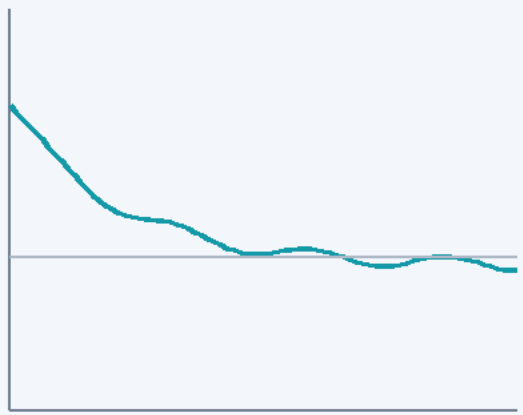
Strong correlation reduces the effective amount of statistical information.

Convergence: look beyond one running average

Analysis & convergence

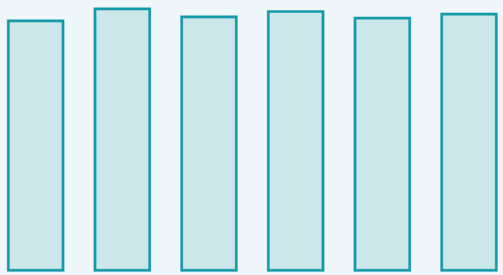
A flat-looking running mean is useful, but it is not sufficient evidence by itself.

Running mean



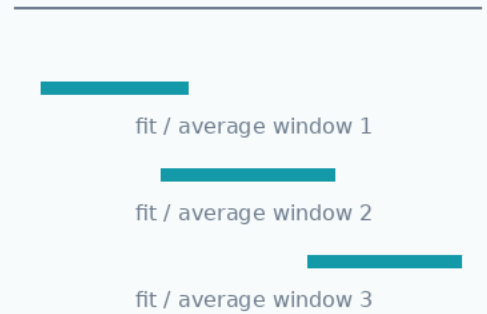
Does the average stabilize?

Block means



Do late blocks agree within fluctuations?

Window sensitivity



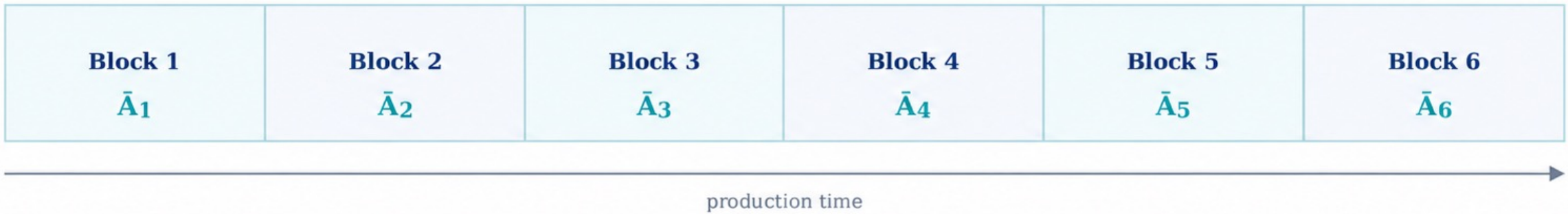
Does the result depend strongly on the chosen window?

Convergence is stronger when several complementary checks tell the same story.

Block averaging: estimating uncertainty from correlated data

Analysis & convergence

Divide the production trajectory into blocks that are long compared with the correlation time.



Average of block means

$$\bar{A} = (1/B)\,\Sigma_\beta\,\bar{A}_\beta$$

Spread among blocks

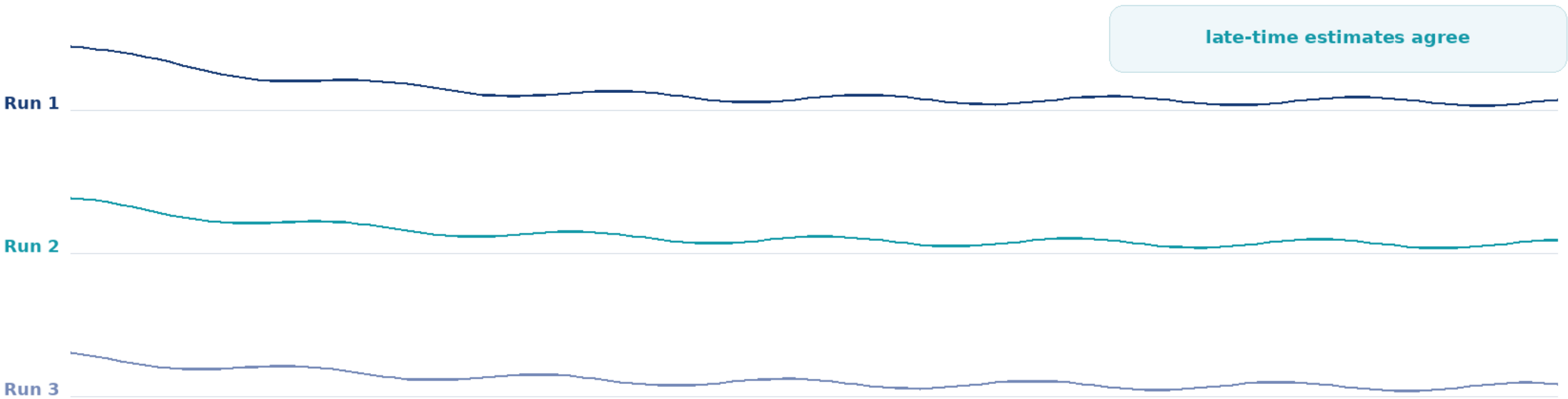
→ **estimate uncertainty**

If blocks are too short, neighboring block means remain correlated and uncertainty is underestimated.

Independent replicas strengthen convergence tests

Analysis & convergence

Different initial velocities or independent starting configurations provide a stronger test than one long trace.



Replica agreement reveals sensitivity to initial conditions and helps expose hidden slow sampling.

Trajectory preprocessing matters

Analysis & convergence

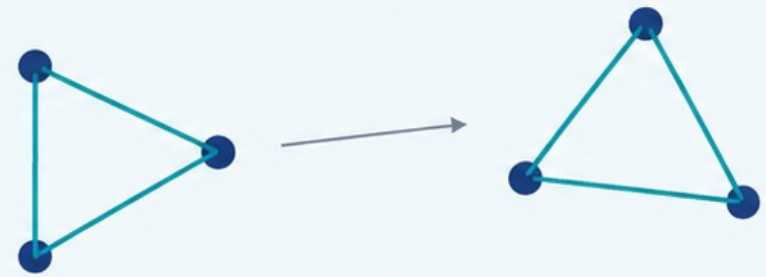
Before calculating an observable, make sure the stored coordinates represent the quantity you intend to measure.

Periodic boundaries: wrapped vs. unwrapped motion



MSD requires a continuous displacement, not jumps caused by wrapping.

Translation / rotation: align only when appropriate



Structural RMSD often uses alignment; diffusion must preserve real translational motion.

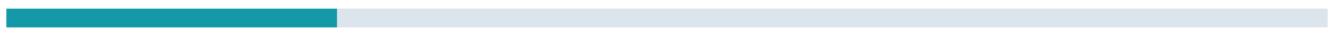
Convergence is observable-specific

Analysis & convergence

A trajectory can be converged for one quantity and still be inadequate for another.

Fast local observable

bond / local energy fluctuations



short

Collective structural observable

RDF, density, conformational population



intermediate

Slow event or transition

barrier crossing, folding, rare transport event



long / possibly ra

There is no single simulation length that guarantees convergence of every observable.

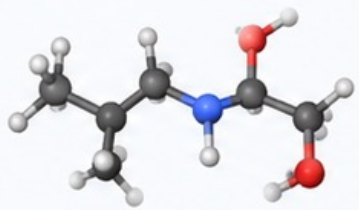
Takeaways: from an MD model to a trustworthy result

Putting it together

Reliable molecular dynamics is a chain of modeling and statistical decisions.

Molecular model

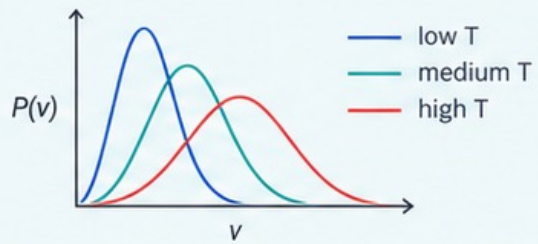
determines the forces



- functional form (bonded, nonbonded)
- parameters (force field)
- level of detail (atomistic, coarse-grained)
- validity and limitations

Ensemble + protocol

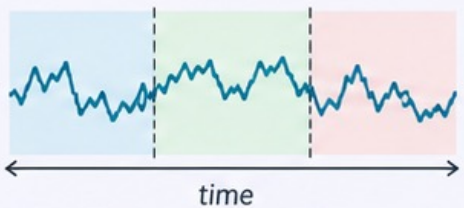
define thermodynamic conditions and preparation



- ensemble (NVE, NVT, NPT)
- thermostat / barostat (if needed)
- system construction and minimization
- equilibration protocol
- time step and constraints
- periodic boundaries and cutoffs

Sampling

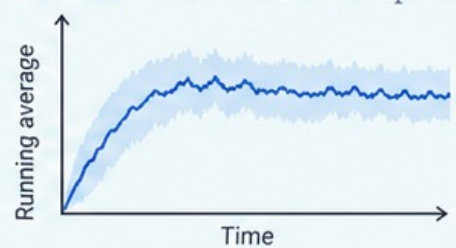
determines which states and timescales are explored



- simulation length
- multiple independent replicas
- relevant timescales for the question
- enhanced sampling (if needed)

Analysis + convergence

determine whether the reported observable is trustworthy



- correct preprocessing (PBC, unwrapping, alignment)
- appropriate observable
- autocorrelation and statistical error (e.g., block averaging)
- convergence checks
- sensitivity to simulation choices
- uncertainty and reproducibility

physical question → **appropriate model** → **validated protocol** → **sufficient sampling** → **converged analysis** → **defensible result**

A trajectory is an intermediate object. The scientific result emerges only after appropriate, validated analysis.